Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1100	**Course Name**	Discrete Structures 1
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 2 / First Year		
3. Course Instructor	Dr. Yassin Mohamed Taher Daada		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Discrete Mathematics and Its Applications, Kenneth H. Rosen, 7th edition, McGraw Hill, 2012 **a. Other supplemental materials:** Discrete Mathematics with Applications, Susanna S. Epp, 4th edition, DePaul University, 2011		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course will introduce the student to a body of mathematical concepts essential for the mastery of some of the higher-level computer science courses. Topics include set theory, functions and relations, propositional and predicate logic, proof techniques, machine level representation of data.		
b. Prerequisites	No pre-requisites		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

<table>
<tr><td>6. Specific goals for the course</td><td colspan="3">a. Specific outcomes of instruction
This course will introduce the student to a body of mathematical concepts essential for the mastery of some of the higher-level computer science courses. The main objective of this course is to develop an understanding of propositional and predicate logics, numerical data representation, sets functions and relations, and be able to write clear, correct proofs.</td></tr>
<tr><td></td><td colspan="3">b. Course Learning Outcomes
Knowledge and Understanding:
Describe the propositional and predicate logics.
Define the following concepts: sets, functions and relations and their properties.
Describe the different numbering systems in computer science.
Skills:
Reconstruct and manipulate different logical expressions using propositional and predicate logics.
Write clear, correct proofs using the presented proofs techniques.</td></tr>
<tr><td>7. Brief list of topics to be covered</td><td colspan="2">Chapter
1. Basic Logic: Propositional logic (cross-reference: Propositional logic is also reviewed in IS/Knowledge Based Reasoning)
2. Sets, Relations, Functions
3. Proof Techniques: Notions of implication, equivalence, converse, inverse, contrapositive, negation, and contradiction
4. Machine Level Representation of Data: Bits, bytes, and words</td><td>No. Weeks
1-4
5-6
6-10
11-12</td></tr>
<tr><td>8. Any other information</td><td colspan="3">Course Assessment</td></tr>
<tr><td></td><td>Activity
Mid Exam
Assignment / Practical
Quizzes
Final Exam
Total</td><td>Assessment Week#
8
As per specified
4-7-9-11
Final week</td><td>and % (Marks)
35%
5%
20 (5*4)%
40%
100%</td></tr>
</table>

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1110	**Course Name**	Introduction to Computer Science
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 1 / First Year		
3. Course Instructor	Dr. Misfer bin Saeed Abdullah Al -Dossary		
Tutorial/Practical Instructor	L; P		
4. Textbook, title, author, and year	**Textbook:** Dale, Nell, and John Lewis. Computer Science Illuminated. 7th ed., Burlington, Massachusetts, Jones & Bartlett Learning, 2020 **a. Other supplemental materials:** None		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This is an introductory course in the first semester of the bachelor programs given by the College of Computer and Information Sciences. It exposes the students to the breadth of the subject. In addition, it gives the students a foundation in problem solving.		
b. Prerequisites	No pre-requisites		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

The purpose of this course is to:

Establish a solid knowledgebase of computer science topics that students will encounter in subsequent courses in their majors.

Prepare the students to subsequent programming courses by the foundations of problem solving.

b. Course Learning Outcomes

Knowledge and Understanding:

Identify the organization of a computer

Distinguish different data representations inside a computer

Recognize the main concepts of computer science topics

Skills:

Analyze the problem and plan the solution by designing the algorithm.

Translate the solution plan to an executable program

Values:

Work effectively in team to accomplish a written/oral report

7. Brief list of topics to be covered

Chapter	No. Weeks
1. Introduction to Course	1
2. Data Representation	1-2
3. Computer Components	3
4. Operating Systems	3
5. Networks, Internet, and WWW	4
6. Artificial Intelligence	5
7. Software Development	5
8. Limitations of Computing	6-7
9. Trending Topics in CS (e.g. IoT)	8-12
10. Algorithms, Problem Solving, and Programming	

8. Any other information

Course Assessment

Activity	Assessment Week#	and % (Marks)
Mid Exam	7	25%
Assignment / Practical	As per specified	10%
Quizzes	As per specified	25%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1120	**Course Name**	Digital Design
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 3 / First Year		
3. Course Instructor	Dr. Alaa eddin Sayed Ahmed mohamed		
Tutorial/Practical Instructor	L; T; P		
4. Textbook, title, author, and year	**Textbook:** Null, Linda, and Lobur, Julia. The essentials of computer organization and architecture, 4th edition, Jones & Bartlett Publishers, 2014. Computer System Architecture, Morris Mano, 3rd edition, Pearson Education, 2007. **a. Other supplemental materials:** Null, Linda, and Lobur, Julia. The essentials of computer organization and architecture, 4th edition, Jones & Bartlett Publishers, 2014. Computer System Architecture, Morris Mano, 3rd edition, Pearson Education, 2007.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course focuses on: Underlying computer hardware including gates, Boolean expressions, combinational circuits analysis and design, flip flops and sequential circuits analysis and design. Understanding the function of a computer system at the machine level, the way the instructions are executed and the way the arithmetic and logic operations are performed. Designing and analyzing the main functional units of a computer such as control unit and data path of a simple CPU.		
b. Prerequisites	CS 1110		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

Analyze and designing combinational and sequential Circuits.
Understand the instruction execution cycle.
Design and analyze the main functional units of a computer such as control unit and data path of a simple CPU.
Write simple assembly computer programs.
Differentiate between different instruction formats.

b. Course Learning Outcomes

Knowledge and Understanding:

Comprehend different logic function minimization techniques (Boolean Algebra and K-Maps).
Understand register transfer language

Skills:

Analyze any digital logic circuit either Combinational or Sequential.
Designing a logic control of a basic computer organization (fetch, decode and execute steps for any instruction).
Summarize how instructions are represented by a timely sequence of micro-operations
Write simple programs at the assembly/machine level.
Use CAD tools for implementation, and simulation to evaluate computer building blocks.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Combinational circuits	1-3
2. Sequential circuits	3-5
3. Digital Logic and Digital Systems	5-6
4. Computer Datapath	7-9
5. Lab Work: Simulating Combinational and Sequential circuits	10-12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Mid Exam	7	30%
Assignment / Project	9	10%
Quizzes	5-10	20%
Final Exam	Final week	40%
Total		100%

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1140	**Course Name**	Introduction to Computer Programming
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 2 / First Year		
3. Course Instructor	Dr. Adel Ibrahim Al -Salem		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Java: How to Program (Late Objects Version), Global Edition, 10/E, Paul Deitel and Harvey Deitel, Pearson, 2015, ISBN-10: 1292019360 • ISBN-13: 9781292019369 **a. Other supplemental materials:** Java: An Introduction to Problem Solving and Programming, 6th Edition By: Walter Savitch		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** The course introduces students to structured programming techniques. Topics include different control statements (sequence, selection, and repetition), fundamental data types, and data structures (arrays). Upon successful completion of the course, students will solve computer problems by using structured programming techniques and adequate tools (text editor, compiler, and debugger).		
b. Prerequisites	No pre-requisites		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

<table>
<tr><td>6. Specific goals for the course</td><td>a. Specific outcomes of instruction
The main purpose of this course is to provide the students the problem solving and programming skills to enable the students to design solutions to nontrivial problems and implement those solutions in Java.</td></tr>
<tr><td></td><td>b. Course Learning Outcomes
Knowledge and Understanding:
Describe the 'uses of' and the ' values that have' a primitive and a compound type.
Recognize the behavior of programs during execution.
Skills:
Develop, test, and debug a program that implements an algorithm that uses the appropriate fundamental programming constructs.
Write programs that use the appropriate data types and structures such as primitive types, arrays, strings, and files.</td></tr>
<tr><td>7. Brief list of topics to be covered</td><td>Chapter

1. Algorithms & Problem Solving
2. Types, Variables, Operators
3. Conditional Control Structures
4. Iterative control Structures
5. Arrays
6. Strings
7. Files</td><td>No. Weeks

1-2
3-4
5-7
8
9
10
11-12</td></tr>
<tr><td>8. Any other information</td><td colspan="2">Course Assessment</td></tr>
<tr><td></td><td colspan="2"><table>
<tr><td>Activity</td><td>Assessment Week#</td><td>and % (Marks)</td></tr>
<tr><td>Mid Exam</td><td>9</td><td>22%</td></tr>
<tr><td>Lab Exams</td><td>4,10</td><td>16%</td></tr>
<tr><td>Lab Assignments</td><td>In lab</td><td>12%</td></tr>
<tr><td>Quizzes</td><td>In-class</td><td>10%</td></tr>
<tr><td>Final Exam</td><td>Final week</td><td>40%</td></tr>
<tr><td>Total</td><td></td><td>100%</td></tr>
</table></td></tr>
</table>

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1141	**Course Name**	Object-Oriented Programming
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 3 / First Year		
3. Course Instructor	Dr. Ahmed Mohamed Ahmed Jad Allah		
Tutorial/Practical Instructor	L; T; P		
4. Textbook, title, author, and year	**Textbook:** Java: How to Program (Early Objects Version), Global Edition, 10/E, Paul Deitel and Harvey Deitel, Pearson, 2015, ISBN-10: 0133807800• ISBN-13: 9780133807806		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course will introduce the student to the concepts of object-oriented programming. Programming topics include data hiding/encapsulation and abstraction using classes and objects, inheritance, polymorphism, generic programming using generic classes and methods, and exception handling.		
b. Prerequisites	CS 1140		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

This course will introduce the student to the concepts of object-oriented programming. The main purpose of the course is to build a strong programming background for future study in computer science. Furthermore, it helps students to develop their problem-solving skills by using the proper logical thinking.

b. Course Learning Outcomes

Knowledge and Understanding:

Identify pitfalls related to inheritance side effects.

Identify the relative strengths and weaknesses among multiple object-oriented designs or implementations for a problem.

Describe the process of analyzing and implementing changes to a large existing code base.

Recall common coding errors that lead to insecure programs (e.g., buffer overflows, memory leaks, malicious code, integer overflows, race conditions) and rewrite a simple program to remove such errors.

Skills:

Design and implement an appropriate object-oriented code for a problem by combining different object-oriented concepts (e.g., encapsulation, polymorphism, and inheritance).

Exploit object-oriented inheritance concept to design inheritance hierarchies and reuse code in subclasses.

Use object-oriented polymorphism to make systems extensible and maintainable.

Create and use generic methods and classes that perform identical tasks on arguments or data fields of different types.

Build GUIs and write event handlers to handle events that are generated by user interactions with GUIs.

Apply the principles of least privilege, fail-safe defaults, and isolation as applied to system design.

Build robust code using exception handling mechanisms.

Apply techniques, coding idioms and mechanisms for creating correct program components that meet programming style standards and achieve desired properties such as reliability, efficiency, and robustness.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Object-oriented design, definition of classes	
2. Object-oriented idioms for encapsulation, using collection classes, iterators, and other common library components.	1-2 3-4
3. Fundamental design concepts and principles	
4. References and aliasing, events and event handlers, canonical uses such as GUIs, mobile devices, robots, servers	5-6 7
5. Generic types (parametric polymorphism)	
6. Program correctness, simple refactoring, Modern programming environments, debugging strategies, documentation and program style	8 9
7. Principle of Least Privilege (PoLP) in OO context, principles of secure design and coding	10-11
8. Coding practices	12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Mid Exam	7	20%
Lab Exams	5-10	30(2*15) %
Quizzes	4-9	10 (2*5) %
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1180	**Course Name**	Computer Science Ethics
2. Credit hours	2	**Contact hours**	30
Level / Year	Level 1 / First Year		
3. Course Instructor	Walaa Nasser Saud Al -Sarami		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** Ethics in Computing, Joseph Migga Kizza, 2016		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course introduces the main concepts, perspectives, and methodological frameworks related to computer science ethics, social issues, and professional practice. The course covers a range of topics, including morality, the law, ethical analysis, professional ethics, anonymity, security, privacy, civil liberties, and intellectual property rights. The course also explores the social context of computing, software issues, computer crimes and cyberbullying, artificial intelligence ethics, virtualization ethics, and cyberspace ethics. Furthermore, the course presents ethical, privacy, and security issues in online social network ecosystems and extends beyond traditional computer networks to address elastic extension.		
b. Prerequisites	No pre-requisites		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

Upon completing this course, students will have developed a comprehensive understanding of the ethical and social implications of computer science and technology, including the importance of professional practice.

b. Course Learning Outcomes

Knowledge and Understanding:

Define the fundamental concepts of computer ethics and professional practice.

Describe a variety of ethical frameworks and relate them to arising ethical issues in computing and professional practice.

Recognize systems issues and their impact on individuals, organizations, and society.

Skills:

Apply best practices for computing ethics within certain context.

Analyze and Explain a variety of malpractices related to computer science field.

Values, autonomy, and responsibility:

Work effectively on teams to accomplish a common goal.

Realize the responsibilities and ethical, legal, security, and social issues associated with the professional practice.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Morality and the Law	1
2. Ethics and Ethical Analysis	2
3. Ethics and the Professions	3
4. Anonymity, Security, Privacy, and Civil Liberties	4
5. Intellectual Property Rights and Computer Technology	5
6. Social Context of Computing	6
7. Software Issues: Risks and Liabilities	7
8. Computer Crimes and Cyberbullying	8
9. New Frontiers for Computer Ethics: Artificial Intelligence, Virtualization, and Cyberspace	9
10. Ethical, Privacy, and Security Issues in the Online Social Network Ecosystems	10
11. Elastic Extension Beyond the Traditional Computer Network	11-12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Mid Exam	6	20%
Assignments	4-7-8-10	20%
Oral Presentations	11	20%
Total	12	40%
		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1201	**Course Name**	Discrete Structures 2
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 4 / Second Year		
3. Course Instructor	Afnan Abdul Rahman Ibrahim Al -Muammar		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Discrete Mathematics and Its Applications, Kenneth H. Rosen, 7th edition, McGraw Hill, 2012 **a. Other supplemental materials:** Introduction to the Theory of Computation, M. Sipser. Course Technology, Boston, MA, Third edition, (2013)		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course will introduce the student to a body of mathematical concepts essential for the mastery of some of the higher-level computer science courses. Topics include counting methods, discrete probability, graph theory, basic automata and computability theory.		
b. Prerequisites	CS 1100		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course	**a. Specific outcomes of instruction** This course will introduce the student to a body of mathematical concepts essential for the mastery of some of the higher-level computer science courses.		
	b. Course Learning Outcomes **Knowledge and Understanding:** Describe the different counting methods. Describe how graph theory and some of its basic algorithms can be used to model and solve problems in the field of computer science. Define basic computing models such as finite state machines and Turing machines. **Skills:** Apply the counting methods and probability to solve real problems Design finite state machines, regular expressions, and context-free grammars to accept a specified language.		
7. Brief list of topics to be covered	Chapter		**No. Weeks**
	1. Counting arguments, permutations and combinations, recurrence		1
	2. Finite probability space, events, axioms of probability and probability measures		2-3
	3. Conditional probability, independence, random variables and probability distribution functions, discrete random variables, expectation, linearity of expectation, variance		4
	4. Graphs, trees		5-7
	5. Automata and Theory of Computations		8-12
8. Any other information	**Course Assessment**		
	Activity	**Assessment Week#**	**and % (Marks)**
	Mid Exam	7	30%
	Assignments	As per specified	10%
	Quizzes	3-5-9-11	20(4*5) %
	Final Exam	Final week	40%
	Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1211	**Course Name**	Design and Analysis of Algorithms
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 5 / Second Year		
3. Course Instructor	Dr. Amal Suleiman Muhammad Al -Saif		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Introduction to Algorithms, Cormen, Leiserson, Rivest, and Stein, 3rd edition, Publisher: MIT Press, 2009, ISBN-13: 978-0262033848, ISBN-10: 9780262033848		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course applies design and analysis techniques to numeric and nonnumeric algorithms which act on data structures. Design is emphasized so that the student feels comfortable with developing new algorithms. Analysis of algorithms is concerned with the resources an algorithm must use to reach a solution. Only theoretical techniques of analysis are covered. Topics include introduction to algorithm, asymptotic complexity, sorting and searching, divide and conquer, data structures, graph algorithms, dynamic programming, and Np-Completeness.		
b. Prerequisites	CS 1242 CS 1201		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

Students should learn the basic skills needed for the Analysis of Algorithms namely, Mathematical Foundations (Growth Functions, Asymptotic Notations, Recurrences). Students should be able to demonstrate how worst-case complexity of an Algorithm is defined. Students should be able to design efficient Algorithms using standard algorithm design techniques. Students should describe and use major algorithmic techniques (sorting, searching, divide-and-conquer, dynamic programming, greedy paradigm, graph algorithms, NP Completeness) and should be able to demonstrate computer science problems for which each technique is suitable.

b. Course Learning Outcomes

Knowledge and Understanding:

Comprehend the principles and mechanisms underlying various kinds of algorithms.
Define how to design and analyze algorithms.
Comprehend the complexity of algorithms.
Define basis of NP-completeness and describe how to solve them using approximation algorithms.

Skills:

Design and analyze algorithms using formal methods and techniques.
Recognize a problem and determine appropriate algorithms to use from set of algorithms learned.
Identify problems that are NP-Complete and determine some approximation methods to solve such problems.

Values:

Work effectively on a team

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Basics of Algorithms, Mathematical Foundations, Insertion Sort, Growth of Functions, Asymptotic notations	1-2
2. Divide-and-Conquer and Recurrence: Merge-sort, Recursion-tree Method, Master Method	3-4
3. Heap-sort, Counting and Radix Sort	5-6
4. Data Structures: Hash Tables, Binary Search Tree	7
5. Graph Theory: Graph Categorization, Graph Terminology, Representation of Graphs, DFS & BFS, Strongly Connected Components, Kruskal Algorithm, Dijkstra's Algorithm	8
6. Dynamic Programming	9
7. Greedy Algorithms	10
8. NP Complete Problems and Solutions using Approximation algorithms.	11
9. Miscellaneous topics: Course Exams, Team Project, and Students Presentations	12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Mid Exam	7	30%
Group-based Project	11	10 %
Quiz	4	10 %
Assignments	During semester	10 %
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1221	**Course Name**	Computer Architecture
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 5 / Second Year		
3. Course Instructor	Dr. Mustafa Al -Sayed Ibrahim		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** The Essentials of Computer Organization and Architecture, Null and Lobur, Jones and Bartlett Pub, 2012. ISBN 0-7637-0444-X. 1. Patterson and Hennessy, Computer Organization and Design: The Hardware and Software Interface. Morgan Kaufmann Pub. 4ed. ISBN: 978-0-12-374493-7. (Used at UW Madison). **a. Other supplemental materials:** Harris and Harris, Digital Design and Computer Architecture. (Used at UC Berkeley). Null and Lobur, The Essentials of Computer Organization and Architecture. Jones Bartlett Pub., 2012. ISBN: 0-7637-0444-X. (Current requirement, get new edition or phase out). Stallings, Computer Organization and Architecture: Designing for Performance. Prentice Hall. 2010. ISBN: 0-1360-7373-5. (Current reference, get new edition or phase out).		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course covers the evolution of computer architecture and the factors influencing the design of hardware of computer architecture and its performance. Topics include instruction set Architectures, performance, performance measures and performance evaluation for computer architectures. Processor micro-architecture, pipelining and pipelining hazards; Cache and main memory organizations; Secondary memory and Redundant Array of Independent Disks (RAID); main concepts of parallel architectures.		
b. Prerequisites	CS 1242 CS 1120 PHY 1104		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

The main purpose of this course is to introduce the students to the computer design process, performance and cost analysis, memory system design issues, instruction set architecture, and pipelining. Demonstrate how to represent floating-point numbers using IEEE floating point representation. Evaluate different computer systems performance using basic performance metrics and Amdahl's Law. Explain instruction set architecture CISC vs RISC. Demonstrate how to map between high-level language patterns into assembly/machine language annotations. Describe how the use of memory hierarchy (cache, virtual memory) is used to reduce the effective memory latency. Compute Average Memory Access Time under a variety of cache and memory configurations and mixes of instruction and data references. Explain basic instruction level parallelism using pipelining and the major hazards that may occur.

b. Course Learning Outcomes

Knowledge and Understanding:

Recognize CISC and RISC architectures and the differences between them.
Understand the MIPS instruction formats and differentiate between the address modes.
Describe variety of memory technologies both internal and external.
Describe the advantages and limitations of RAID architectures.

Skills:

Apply IEEE floating point representation for representing floating-point numbers.
Compute Average Memory Access Time under a variety of memory levels.
Evaluate different computer systems' performance using basic performance metrics and Amdahl's Law.
Explain basic instruction level parallelism using pipelining and the major hazards that may occur.
Demonstrate how to map between high-level language patterns into assembly/machine language annotations.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Introduction and Performance (Patterson ch1). Introduction, History, Performance measure	1
2. Computer Arithmetic. Review Data representation, Floating point, Some computer arithmetic.	2-3
3. Instruction Sets. Components of an instruction set, Addressing modes, RISC and CISC.	4-5
4. Processor Organization: MIPS.	
5. Memory and types of memories, Memory Hierarchy, Caches (review cache mapping, Replacement algorithms, Access time, hit ratio, levels), External memory (RAID).	6-7
6. Pipelining. Basic pipelined data path and control, pipeline performance; Data dependences, data hazards, bypassing, code scheduling, Branch hazards, delayed branches, branch prediction.	8-10
7. Miscellaneous topics: Course Exams, Team Project, and Students Presentations	11-12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Mid Exam	6	30%
Project/Assignments	All over semester	10 %
Quizzes	4-9	20 %
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1222	**Course Name**	Operating Systems
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 6 / Second Year		
3. Course Instructor	Dr. Haifa Hamad Abdul Rahman bin Qasim		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Operating System Concepts, 10th Edition, Abraham Silberschatz, Peter B. Galvin, Greg Gagne, 2018. (Main textbook) Modern Operating Systems, 4th Edition, Andrew S. Tanenbaum, Herbert Bos, 2014. **a. Other supplemental materials:** Understanding Operating Systems. 8th Edition. Ann McHoes and Ida M. Flynn. Cengage Learning. 2017		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** Operating systems are an essential part of any computer system. Similarly, a course on operating systems is an essential part of any computer-science education. This course aims to provide a clear description of the theoretical concepts that underlie operating systems. It also aims to familiarize students with the practical side of the OS by programming and simulating different aspects Threading, scheduling, Synchronization, memory management, etc.).		
b. Prerequisites	CS 1221		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

The main purpose for this course is to introduce students to the operating systems concepts, principles, and design. The focus is on presenting the computing systems layers and explaining different methods of handling problems such as deadlocks through these layers. Another emphasis is on identifying and comparing different communication and synchronization mechanisms that can be employed to realize concurrent systems and shared resources. These mechanisms deal with multi threads systems and include important topics such as critical section problems and semaphore. Also, Processor scheduling algorithms for both preemptive and non-preemptive are introduced. Another emphasis is to introduce to students the main principles of memory and virtual memory management techniques such as segmentation and paging. The goal is to provide the students with a set of important concepts to acquire the ability to handle, design and use operating system-based techniques within the computer science area context.

b. Course Learning Outcomes

Knowledge and Understanding:

Define the operating systems concepts, principles, and design.

Understand the principles of process and threads management in OS environment such process communication, process synchronization, handling deadlocks and scheduling shared resources.

Recognize the principles of memory management such as allocation methods of main memory (segmentation and paging) and virtual memory (different techniques) management.

Skills:

Compare the different communication and synchronization mechanisms that can be employed to realize concurrent systems and shared resources.

Evaluate different processor scheduling algorithms for both preemptive and non-preemptive.

Apply different inter-process and multi threads communication and synchronization mechanisms, such as critical section problem and semaphore.

Values:

Work in teams and learn how to manage the time slot allocated to the project.

Know how to present the implemented projects

7. Brief list of topics to be covered

Chapter	No. Weeks
Introduction into OS	1
1. Processes	2-3
2. Threads	4
3. Synchronization Tools	5
4. Synchronization Examples	5
5. CPU Scheduling	6
6. Deadlock	6
7. Main Memory	7-8
8. Virtual Memory	9-11
9. Miscellaneous Topics: Project submission and presentation	12

8. Any other information

Course Assessment

Activity	Assessment Week#	and % (Marks)
Mid Exam	7	25%
Project and presentation	10	15 %
Quizzes	4-8	10 (2*5) %
Assignments	4-8	10%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1242	**Course Name**	Data Structures
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 4 / Second Year		
3. Course Instructor	Hedi Suleiman		
Tutorial/Practical Instructor	L; T; P		
4. Textbook, title, author, and year	**Textbook:** Data Structures and Algorithm Analysis in Java, Third Edition, Mark Allen Weiss, Publisher: Pearson, 2012, ISBN-13: 978-0132576277, ISBN-10: 0132576279. **a. Other supplemental materials:** Java How To Program-Early Objects, Third Edition, Deitel, P. and Deitel, H., Publisher: Prentice Hall Press, 2015, ISBN-10: 0132576279, ISBN-13: 978-0132576277. Open Data Structures (in Java), Pat Morin, Publisher: UBC Press; 31st ed. Edition, 2013, ISBN-10: 1927356385, ISBN-13: 978-1927356388. Data Structures and Algorithms in Java, Robert Lafore, 2nd Edition, Publisher: Sams Publishing, 2012, ISBN-10: 0672324539, ISBN-13: 978-0672324536. Java Structures: Data Structures in Java for the Principled Programmer, Duane A. Bailey, Second Edition, Publisher: McGraw-Hill Science/Engineering/Math, 2007, ISBN-10: 0072399090 ISBN-13: 978-0072399097		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course provides the students with understanding of the concepts on data representation and organization used in development of computer applications. This includes basic and advanced data structures such as Linked Lists, Stacks, Queues, Binary Trees, Binary Search Trees, Hash Tables and Graphs. The students are trained to use the programming concepts such as Abstraction and encapsulation to implement the data structures using Java language. The course will also introduce basic knowledge of algorithm analysis and, space and time complexity. Various sorting and searching algorithms will be introduced as well.		
b. Prerequisites	CS 1141 CS 1100		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

The main purpose of this course is to provide students the knowledge of data representation and their organization with respect to concepts of time and space complexity.

b. Course Learning Outcomes

Knowledge and Understanding:

Describe the basic and advanced data structures and their applications for real world software systems

Skills:

Compute the time complexity for small programs.

Implement the appropriate concepts related to data structures for solving a particular problem.

Design and implement the appropriate Classes/Interfaces/Functions for a small to medium-level problem with space/time efficiency in mind.

Apply basic and advanced sort algorithms to solve a problem.

Apply a variety of search algorithms to solve a problem.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Review of Encapsulation, Inheritance, and Simple Data Structures in Java. This will address the basic Java classes and interfaces. Introduction to Algorithm Analysis.	1
2. Linked Lists: Single linked list, Double link list, and Circular linked list.	2-3
3. Stacks: Basic concepts and implementation, RPN (Postfix expression) evaluation.	4
4. Queues: Basic concepts and implementation, circular array implementation using arrays.	5
5. Trees: binary tree, binary search tree, expression trees.	6-7
6. Searching: Linear search and Binary Search.	8
7. Hash Tables: Hash function implementation and Collision concept.	9
8. Graph Data Structures: Shortest Path and Minimum Spanning Tree.	10
9. Sorting: Bubble Sort, Insertion Sort, Quick Sort, and Selection Sort	11-12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Mid Exam	7	20%
Lab Quizzes	4-8	15 %
Quizzes	3-9	15 %
Lab work: Assignments	2-10	10%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1250	**Course Name**	Software Engineering 1
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 5 / Second Year		
3. Course Instructor	Dr. Aram Saleh Hamad Al -Sidrani		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** BKCASE Editorial Board. 2017. The Guide to the Systems Engineering Body of Knowledge (SEBoK) Timothy C. Lethbridge and Robert Laganière; Object-Oriented Software Engineering: Practical Software Development using UML and Java Second Edition Software Engineering: A Practitioner's Approach 9th Edition by Roger S. Pressman Software Engineering. Ninth Edition. Ian Sommerville. Addison-Wesley.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course provides the students with understanding to methods & techniques for software development: Software development activities (requirements specification, design, architecture implementation, validation, testing, and quality assurance) and Software development models (Plan-driven, Incremental, Prototype and Agile, etc.).		
b. Prerequisites	CS 1242		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

Course of Software Engineering aims to develop the thorough knowledge of software development process in students. Course focuses on different phases of Software Development Life Cycle (SDLC) and gives students an insight of tools, techniques and methodologies related to each phase. With the help of homework assignments, and hands on experience for completing different tasks in software development would make students skilled and capable to develop software independently and with teams.

b. Course Learning Outcomes

Knowledge and Understanding:

Identify the basic components of different process models in software development life cycle and realize the impact of the choice of software process model.
Describe the software validation and verification and the different types of testing.
Recognize the basic activities and associated risks for software project management.
Describe the change management process and role of refactoring in it along with identifying the principal issues associated with software evolution.
Define the concepts of software reliability, software quality, and software process improvement models and approaches.

Skills:

Analyze, document, manage and trace different functional and non-functional requirements for the software system.
Evaluate the stakeholder's position regarding different technical decisions.
Develop & design appropriate models for the structure and behavior of software products from their requirements specifications.

Values:

Work effectively within a group to develop a software project.
Recognize ethical and professional responsibilities and act accordingly during the development of software system.

7. Brief list of topics to be covered

Chapter	No. Weeks
1. Introduction to software process models	1-2
2. Software process models	3-4
3. Describing functional requirements using	5-6
4. Vulnerabilities in Information Management Systems	7
5. Communicating professionally with stakeholders.	8
6. Ethical theories and decision-making	9
7. System design principles	10
8. Software development in the context of large Characteristics	11-12

8. Any other information

Course Assessment

Activity	Assessment Week#	and % (Marks)
Mid Exam	9	20%
Group Project (part 1)	4	15 %
Group Project (part 2)	7	20 %
Assignments	3-5-7-9-11	5%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1251	**Course Name**	Software Engineering 2
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 6 / Second Year		
3. Course Instructor	Dr. Imran Quraishi		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Sommerville, I. (2016) Software Engineering. 10th Edition, Pearson Education Limited, Boston. BKCASE Editorial Board. 2017. The Guide to the Systems Engineering Body of Knowledge (SEBoK) **a. Other supplemental materials:** Timothy C. Lethbridge and Robert Laganière; Object-Oriented Software Engineering: Practical Software Development using UML and Java Second Edition. Software Engineering: A Practitioner's Approach 9th Edition by Roger S. Pressman.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course cover the remaining topics in software engineering for the software development life cycle (SDLC) along with more detail practical topics in requirements, design, verification and validation, implementation, and evaluation with focus on software quality assurance, project management and maintenance. Also, introduced different advanced topics in software engineering, such as software tools where there are used in different software development process to evaluate the software product and its quality.		
b. Prerequisites	CS 1250		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

c. Specific outcomes of instruction

This course is a complementary course for Software Engineering 1 course. Its allow students to deep understanding for theoretical and practical issues in building CS projects before start working with the graduate project.

d. Course Learning Outcomes

Knowledge and Understanding:

Describe different aspects regarding the code, such as construct, execute, and debug the code of a program to make it secure, robust, and free of vulnerabilities.

Analyze a use case or similar description of some behavior with focusing on the critical components of a data model to identify software requirements.

Recognize the basic activities and associated risks for software project management.

Describe the change management process and role of refactoring in it along with identifying the principal issues associated with software evolution.

Define the concepts of software reliability, software quality, and software process improvement models and approaches.

Skills:

Analyze the reliability and impact of software architectures selection on the design of a simple system.

Develop & design software products that meets quality requirements.

Implement software tools in support of the development, configuration, and testing of a software product.

Apply formal specification, using a common formal specification language, program assertions, and formal analysis techniques to software designs and programs.

Apply software tools in support of the development, configuration, and testing of a software product.

values:

Work effectively on teams to accomplish a common goal.

Recognize the need for and an ability to engage in continuing professional development.

Realize professional, ethical, legal, security and social issues and responsibilities.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Program correctness	1
2. Security by design	2
3. Object-oriented analysis and design	3
4. Software fault tolerance techniques and model	4
5. Software configuration management and version control	5-6
6. Coding practices	7-8
7. Testing fundamentals and test-case generation &Unit testing	9-10
8. Team organization and decision-making	11-12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Mid Exam	8	20%
Project	3.5.7.10	20 %
Assignments	3.5.7.10	20 %
Final Exam	Final week	40%
Total		100%

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1343	**Course Name**	Compilers
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 9 / Fourth Year		
3. Course Instructor	Dr. Manal Khaled Nayef Al -Sabhan		
Tutorial/Practical Instructor			
4. Textbook, title, author, and year	**Textbook:** Principles of compilers: A New Approach to Compilers Including the Algebraic Method, Su, Yunlin, Yan, Song Y, Springer, 2011, ISBN-10: 3642208347, ISBN-13: 978-3642208348 **a. Other supplemental materials:** Engineering a Compiler by Cooper and Torczon Modern Compiler Implementation in Java (Second Edition) by Andrew Appel, Cambridge, 2002, ISBN-10: 052182060X, ISBN-13: 978-0521820608 Online: Basics in Compiler Design by Torben Mogensen YouTube channels that explain the compliers phases with more tutorials Flex and Bison		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course aims to cover the main technologies associated with implementing programming languages: scanning, lexical analysis, syntax analysis, type checking, and brief overview of optimization, run-time data organization and code-generation.		
b. Prerequisites	CS 1251 CS 1211		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction (course objective)

The course describes definition and using of the main compiler construction phases specially lexical analyzer using Finite Automata theory, syntax parser using Context-Free-Grammar, semantic analyzer focusing on type and scope checking. Student should define and resolve ambiguity in regular expressions, parsing and type and scope checking. The course gives also overview on how to generate a machine code and optimize it, and memory allocation operations.

b. Course Learning Outcomes

Knowledge and Understanding

Recognize the principles and mechanisms underlying implementation of compliers.
Define using regular expression on recognizing types of tokens.
Define the operations of semantic analysis and using inference rules for type checking.

Skills :

Construct DFA and NFA for a regular expression.
Generate unambiguous Context-free-grammars for syntactic parsing.
Differentiate between syntax parsing techniques top-down and bottom-up and use them in the construction of parsers.
Distinguish between static and dynamic scope checking, compile time and run time operations.

Values:

Work effectively in groups and exercise leadership when appropriate

7. Brief list of topics to be covered

Topics	No . Weeks
1- Introduction to Compilers	1
2- Lexical Analyzer	2-4
3- Syntax Analyzer	5-7
4- Semantic Analyzer	8-11
5- Project discussion	12

8. Any other information

Course Assessment

Activity	Assessment Week	# and % (Marks)
Quizzes	5-9	14%
Assignment	10	6%
Projects	10	20%
Midterm	7	20%
Final Exam	Final week	40%

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COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1323	**Course Name**	Distributed Systems
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 9 / Third Year		
3. Course Instructor	Dr. Manan Muhammad Zaid Al -Muslim		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** George Coulouris, Jean Dollimore & Tim Kindberg: Distributed Systems: Concepts and Design, 5th ed, 2011, Addison-Wesley. **a. Other supplemental materials:** Andrew S. Tanenbaum & Maarten van Steen: Distributed Systems: Principles and Paradigms, 2nd ed, 2017, Pearson Prentice Hall Doren L. Galli: Distributed Operating Systems, 1999, Prentice Hall.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** The course aims to provide an understanding of the essential principles of distributed systems. Topics coved include distributed computing definition and challenges, Architectural Models, Interposes communication, Distributed Objects and Remote Invocation, Operating System Support, Web services and distributed algorithms.		
b. Prerequisites	CS 1330		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

Know the principles of Distributed systems and its key performance challenges.
Provide the students with a set of tools to implement the characteristics of distributed models to be able to show their progress guarantees and correctness with respect to safety or liveness properties. Focus on showing methods of conducting different a performance experiment on a layered system to determine the effect of a system parameter on figure of system performance. Understand synchronization constructs such as Clock synchronization, Mutual exclusion and election algorithms show its affect on the distributed algorithm's communication costs.

b. Course Learning Outcomes

Knowledge and Understanding:

Identify core concepts and techniques of distributed systems.
Define process concept, synchronization and communication.
Understand the concept of distributed concurrency control and distributed algorithms.

Skills:

Create a performance experiment on a layered system to emphasize the concept of applying Distributed Objects and Remote Invocation.
Analyze the characteristics of particular formal distributed models in order to be able to show their progress guarantees and correctness with respect to safety or liveness properties.
Provide experience in the analysis implementation of typical algorithms used in distributed systems.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Characterization of Distributed Systems	1
2. Distributed Systems Models	2-3
3. Inter-process Communication	4-5
4. Distributed Objects and Remote Invocation	6-7
5. Operating System Support	8
6. Web services	9
7. Examples of Distributed Systems	10-11
8. Miscellaneous topics: Course Exams, Team Project, and Students Presentations	12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Mid Exam	7	25%
HomeWorks	Every 3 weeks	15 %
Quizzes	5.11	20 %
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1330	**Course Name**	Computer Networks
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 7 / Third Year		
3. Course Instructor	Dr. Ibrahim Abdel -Moneim Abdel -Majid		
Tutorial/Practical Instructor	L; P		
4. Textbook, title, author, and year	**Textbook:** J. Kurose and K. Ross, Computer Networking: A Top-Down Approach, 7th Edition ed., Pearson, 2016. Opnet Laboratories could be selected from: L. L. Peterson and B. S. Davie, Computer Networks: A Systems Approach, 5th Edition ed., Morgan Kaufmann, 2011. **a. Other supplemental materials:** E. Aboelela, Network Simulation Experiments Manual, 3rd Edition ed., The Morgan Kaufmann Series in Networking., 2011.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** The course introduces fundamental concepts in the design and implementation of computer communication networks and their protocols. This includes layered network architectures, applications, transport, congestion, routing, data link protocols, and local area network concepts. An emphasis will be placed on the protocols used in the Internet. A top-down approach will be emphasized during the course starting from the application layer down to the data link layer.		
b. Prerequisites	CS 1222		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

<table>
<tr><td>6. Specific goals for the course</td><td colspan="3">a. Specific outcomes of instruction (course objective)
By the completion of the course the student should be able to:
This course aims to introduce students to the internet organization, network components and the network layers services and protocols.</td></tr>
<tr><td></td><td colspan="3">b. Course Learning Outcomes
Knowledge and Understanding
Describe wired and wireless networks organization, the network layered structure and security issues related to different layers.
List and define some internet protocols such as IP, TCP, UDP, HTTP, DNS, ICMP.
Recognize the processes of circuit and packet switching, reliable data transfer and congestion control and common multiple access approaches.

Skills:
Apply algorithms and mathematical computations related to computer networks such as routing algorithms, delays computation, error detection and correction methods and subnetting.
Design and implement network components such as client-server socket-based application, programs using methods of layering, or a simple protocol.</td></tr>
<tr><td>7. Brief list of topics to be covered</td><td colspan="2">Chapter</td><td>No. weeks</td></tr>
<tr><td></td><td colspan="2">1. Computer Networks and the Internet
2. Application Layer
3. Transport Layer
4. Network Layer
5. Data Link Layer and Local Area Networks
6. Wireless and Mobile Networks
7. Miscellaneous topics: Course Exams, Team Project, and Students Presentations</td><td>1-2-3
3-4
4-5-6
7-8-9
9-10-11
11
12</td></tr>
<tr><td>8. Any other information</td><td colspan="3">Course Assessment</td></tr>
<tr><td></td><td>Activity</td><td>Assessment Week# and % (Marks)</td><td></td></tr>
<tr><td></td><td>Quiz</td><td>4, 10</td><td>5%</td></tr>
<tr><td></td><td>Midterm Exams</td><td>6, 12</td><td>40%</td></tr>
<tr><td></td><td>Lab Assignment</td><td>9</td><td>5%</td></tr>
<tr><td></td><td>Project</td><td>15</td><td>10%</td></tr>
<tr><td></td><td>Final Exam</td><td>Final week</td><td>40%</td></tr>
<tr><td></td><td>Total</td><td></td><td>100%</td></tr>
</table>

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1344	**Course Name**	Application Development
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 9 / Third Year		
3. Course Instructor	-		
Tutorial/Practical Instructor	L; P		
4. Textbook, title, author, and year	**Textbook:** Internet & World Wide Web, How to Program, Paul Deitel, Harvey Deitel, and Abbey Deitel, 5th edition, ISBN-10: 0-13-215100-6, ISBN-13: 978-0-13-215100-9, Prentice Hall, 2011. Griffiths, D., & Griffiths, D. (2017). Headfirst Android Development: a brain-friendly guide. " O'Reilly Media, Inc.". Game Design Workshop: A Playcentric Approach to Creating Innovative Games, Third Edition, 2014, Tracy Fullerton **a. Other supplemental materials:** Learning Web Design: A Beginner's Guide to XHTML, Style Sheets, and Web Graphics, Niederst, O'Reilly, 2007. Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker 1st ed, 2020 Murphy, M. L. (2008). The busy coder's guide to Android development. United States: CommonsWare, 2008 Schell, J. (2014). The Art of Game Design: A book of lenses. AK Peters/CRC		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course explains to students the concepts of web, mobile, and game applications development.		
b. Prerequisites	CS 1352		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

<table>
<tr><td>6. Specific goals for the course</td><td colspan="2">a. Specific outcomes of instruction (course objective)
By the completion of the course the student should be able to:
This course aims to:
Get introduced to the life cycle of application developments
Get introduced to design and data exchange through a web application
Get introduced to design and data exchange through a mobile application
Get introduced to design and data exchange through a game application</td></tr>
<tr><td></td><td colspan="2">b. Course Learning Outcomes
Knowledge and Understanding
Understand the life cycle of application developments
Describe the usage of tools and languages for application design, client-side scripting, server-side scripting, and data exchange
Skills:
Design application UI based on standard guidelines
Develop an application for a specific usage based on software requirements analysis
Clearly communicate ideas and solutions of programming problems to others
Values
Work effectively in teams to develop Web-based solutions</td></tr>
<tr><td>7. Brief list of topics to be covered</td><td>Chapter</td><td>No. weeks</td></tr>
<tr><td></td><td>1. Introduction to application development
2. Basics, structure, and lifecycle of web applications
3. Basics, structure, and lifecycle of mobile and smart devices applications
4. Basics, structure, and lifecycle of Gaming applications
5. UI Design for web, mobile, and game applications
6. Multi-tier architecture (MVC, MVVM)
7. Web, mobile, and game application trends in cutting-edge technology
8. Misc. Topics: Written Exams and Students Projects</td><td>1
2-3-4
4-5-6
6-7-8
9
10
11
12</td></tr>
<tr><td>8. Any other information</td><td colspan="2">Course Assessment</td></tr>
<tr><td></td><td>Activity
Quizzes (2)
Midterm Exam (1)
Project
Final Exam
Total</td><td>Assessment Week# and % (Marks)
During semester 20%
During semester 20%
End of semester 20%
Final week 40%
100%</td></tr>
</table>

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1352	**Course Name**	Human-Computer Interaction
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 8 / Third Year		
3. Course Instructor	Dr. Sarah Muhammad Ibrahim Al -Hassan		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** Alan Dix, Janet E. Finlay, Gregory D. Abowd, Russell Beale. "Human-computer interaction.", 3rd Edition, 2009. **a. Other supplemental materials:** Jenny Preece, Helen Sharp, Yvonne Rogers, "Interaction Design: Beyond Human-Computer Interaction", 4th Edition, 2015 (http://www.id-book.com/)		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course presents different interaction models, frameworks and styles. Moreover, it includes the interaction design process and highlights the range of design rules that can help to increase the usability of software products. In addition, it includes the evaluation techniques under two broad headings: expert analysis and user participation. Furthermore, it discusses how to design a system to be universally accessible, regardless of age, gender cultural background, or ability.		
b. Prerequisites	CS 1370		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

<table>
<tr><td>6. Specific goals for the course</td><td colspan="3">a. Specific outcomes of instruction (course objective)
By the completion of the course the student should be able to:
This course provides an introduction to the field of Human computer Interaction (HCI). Therefore, the course provides an overview about the fundamental components of an interactive system which include the human, the computer system itself and the nature of the interaction</td></tr>
<tr><td></td><td colspan="3">b. Course Learning Outcomes
Knowledge and Understanding
Define the functional capabilities and limitations of humans and of computing systems in the interaction process
Recognize and outline the advanced aspects, principles, tools and techniques related to Human Computer Interaction (HCI)
Skills :
Demonstrate the ability to apply the HCI design principles in practical problems
Evaluate appropriate evaluation techniques in HCI and develop an understanding of universal design
Values:
Learn to work effectively in teams to design\evaluate the usability of interactive systems</td></tr>
<tr><td>7. Brief list of topics to be covered</td><td colspan="2">Chapter</td><td>No. weeks</td></tr>
<tr><td></td><td colspan="2">1. Input/output channels.
2. Actual physical controls
3. Model of interaction
4. Interaction techniques and paradigms
5. Design definition and process
6. The software life cycle
7. Principles to support usability
8. Design implementation
9. Interface evaluation and goals</td><td>1
2
3
4
5-6
6
7-8
8-9-10
10-12</td></tr>
<tr><td>8.Any other information</td><td colspan="3">Course Assessment</td></tr>
<tr><td></td><td>Activity
Project
One Midterm exam
Quizzes
Final exam
Total</td><td colspan="2">Assessment Week# and % (Marks)
11 10%
9 30%
6-8 20%
Final week 40%
100%</td></tr>
</table>

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1372	**Course Name**	Information Security
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 8/ Third Year		
3. Course Instructor	Dr. Abeer Nafjan Muhammad Al -Nafjan		
Tutorial/Practical Instructor	L; P		
4. Textbook, title, author, and year	**Textbook:** William Stallings, Cryptography and Network Security Principles and Practices, Sixth Edition, Pearson, 2013 William Stallings and Lawrie Brown, Computer Security: Principles and Practice, Fourth Edition, Pearson, 2017 **a. Other supplemental materials:** Mike Speciner and Radia Perlman, Network Security: Private Communication in a Public World, 2nd Edition, 2002.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course will provide students with an academic overview of Information security covering its main domains with more focus on cryptography, network security, and governance. By completion of this course, students should appreciate the significance of information security in the IT realm in general, and be able to demonstrate in depth knowledge in crypto and network security in specific.		
b. Prerequisites	CS 1330, MATH 1227, CS 1180		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction (course objective)

This course aims to provide students with knowledge and practical skills related to Information security which became one of the core domains in computer science. Students have to cover concepts and implement best practices of Information security across all domains of IT. This course also incorporates the use of virtual-lab environment to enable students with suitable training exercises.

b. Course Learning Outcomes

Knowledge

Describe the concepts of authentication, authorization, access control, the concept of trust and trustworthiness.

Define the concepts of risk, threats, vulnerabilities and attack vectors (including the fact that there is no such thing as perfect security).

Identify the limitations of malware countermeasures and how Denial of Service attacks can be identified and mitigated

Define the following terms: cipher, cryptanalysis, cryptographic algorithm, and cryptology, and describe the two basic methods (ciphers) for transforming plain text in cipher text

Skills:

Explain likely attacker types against a particular system such as, instances of social engineering attacks and Denial of Service attacks.

Recognize the appropriate defense mechanism(s) and its limitations given a network threat

Explain how public key infrastructure supports digital signing and encryption and discuss the imitations/vulnerabilities.

7. Brief list of topics to be covered

Topics	No . Weeks
1. CIA (Confidentiality, Integrity, Availability)	1
2. Basic Cryptography Terminology	2-4
3. Network specific threats and attack types	5-7
4. Defense in depth	8
5. Complete mediation	9
6. Attacker goals, capabilities, and motivations	10
7. Attacks on privacy and anonymity	11-12

8. Any other information

Course Assessment

Activity	Assessment Week	# and % (Marks)
Lab exams/quizzes	As per preference	10%
Project	11	10%
Midterm Exams	6	40%
Final	Final week	40%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1360	**Course Name**	Artificial Intelligence
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 7/ Third Year		
3. Course Instructor	Dr. Doha Ahmad Bin Hady Noureldeen		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Russel & Norvig : Artificial Intelligence : a Modern Approach, 3rd edition, 2010. David L. Poole (Author), Alan K. Mackworth: Artificial Intelligence: Foundations of Computational Agents 1st Edition, 2010. **a. Other supplemental materials:** None		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course covers in depth the central topics of Artificial Intelligence. Namely, a large part of this course will be devoted to searching strategies in problem solving, knowledge representation and reasoning. Some selected topics will also be covered such as reasoning under uncertainty and machine learning		
b. Prerequisites	CS 1211 MATH 1227 STA 1011		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction (course objective)

By the completion of the course the student should be able to:

Students will learn the basic skills needed in AI, namely, Searching, Reasoning, Learning, and Knowledge Representation & Intelligent Agent. In addition, students will be able to use, develop, and maintain intelligent systems in real world applications. Students should be familiarized with advanced topics in AI using appropriate tools & Programming languages. The students will have clear understanding about the state of the art in AI, the key issues in AI and which way AI is headed.

b. Course Learning Outcomes

Knowledge and Understanding

Define specific problem of AI, complex concepts and intelligent strategies.

Describe the representation and reason about knowledge in a computer

Recognize the technical issues that arise within Artificial Intelligence

Skills

Design intelligent and traditional computer-based systems, using formal design procedures.

Formulate problems requiring a combination of techniques from Artificial Intelligence.

Develop abstract representations and appropriate solutions for real problems

Values

Work effectively in group projects

7. Brief list of topics to be covered

Chapter	No. weeks
1. Introduction (Artificial Intelligence & Agents)	1
2. States & Search (uninformed, informed, Games: adversarial search)	2-3-4
3. Features & Constraints	5
4. Knowledge Based Systems	6-7-8
5. Reasoning under Uncertainty	8-9-10
6. Learning	11
7. Miscellaneous topics: Course Exams, Team Project, and Students Presentations	12

8. Any other information

Course Assessment

Activity	Assessment Week#	and % (Marks)
Quizzes (2)	3, 5	20%
Midterm	7 or 8	25%
Project	11	15%
Final	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1361	**Course Name**	Machine Learning
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 8/ Third Year		
3. Course Instructor	Dr. Asmaa Badr Al -Saleh		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Machine Learning, by Tom Mitchell, McGraw-Hillm,1997 Understanding Machine Learning: From Theory to Algorithms, by Shai Shalev-Shwartz and Shai Ben-David, Cambridge University, 2014. **a. Other supplemental materials:** Murphy, Kevin P. Machine learning: a probabilistic perspective. MIT press, 2012. Research papers, Technical Reports.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** Machine learning uses interdisciplinary techniques such as statistics, linear algebra, optimization, and computer science to create automated systems that can sift through large volumes of data at high speed to make predictions or decisions without human intervention. Based on fundamental knowledge of computer science principles and skills, probability and statistics theory, and the theory and application of linear algebra, this course provides a broad introduction to machine learning models. It will familiarize students with a broad cross-section of predication and clustering models and algorithms for machine learning and prepare students for industry application of machine learning techniques.		
b. Prerequisites	CS 1360		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction (course objective)

By the completion of the course the student should be able to:

Upon completion of this course, students will have gained knowledge of the fundamentals of machine learning concepts and theory that enable them to apply them on real use cases.

b. Course Learning Outcomes

Knowledge and Understanding

Recognize the concepts and theory of machine learning.

Identify different techniques and tools of machine learning and their domains of application.

Skills

Apply algorithmic principles of machine learning in the modelling and design of computer-based systems.

Develop various machine learning models in a range of real-world applications.

Evaluate different machine learning models.

Values

Work effectively on teams to accomplish a common goal.

7. Brief list of topics to be covered

Chapter	No. weeks
1. Concept Learning and the General-to-Specific Ordering	1
2. Decision Tree	1-2
3. Bayesian Learning	2-3
4. Instance Based Learning, Nearest Neighbor Methods	3-4
5. Support Vector Machine	4-5
6. Linear and Logistic Regressions	6
7. Ensemble Learning	7
8. Unsupervised Learning; Clustering	8-9
9. Artificial Neural Networks, Deep Learning	10-11
10. Misc. Topics: Written Exams and Students Project	12

8. Any other information

Course Assessment

Activity	Assessment Week# and % (Marks)	
Project	As per preference	20%
Midterm	8	20%
Quizzes	As per preference	10%
Assignment (max. 2)	As per preference	10%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1370	**Course Name**	Principles of Database
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 7/ Third Year		
3. Course Instructor	Dr. Mubarak Al -Bathan		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Fundamentals of Database Systems, Elmasri and Navathe, 7th edition, Addison-Wesley, 2016, ISBN-13: 978-0133970777 ISBN-10: 0133970779 **a. Other supplemental materials:** Database System Concepts, Silberschatz, Korth, and Sudarshan, 6th edition, McGraw-Hill, 2011, ISBN-13: 978-0073523323, ISBN-10: 0073523321. Data Modeling and Database Design, Umanath, 2nd edition, 2015, ISBN-13: 978-1285085258, ISBN-10: 1285085256.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course introduces the basic concepts of databases and database management systems. And provides the students with a clear understanding of the different steps in the data design validation and implementation. Topics of this course should complement the topics covered in the Software Engineering course which focuses on the functional design		
b. Prerequisites	CS 1251		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction (course objective)

By the completion of the course the student should be able to:

The aim of this course is to introduce the concept of database, its purpose and advantages to the students. Relational model, normalization, ER diagrams, relational databases, relational algebra will be taught as well as the use of query languages such as SQL in a programming environment, in order to enforce theoretical and practical understanding of the course to build a complete database system.

b. Course Learning Outcomes

Knowledge and Understanding

Outline the main benefits of and actors in a database system.

Recognize the components of a conceptual model namely the ER model.

Recall the syntax of the main SQL statements.

State the normal forms a database design should comply with.

Skills :

Design the ER model from a description.

Create a relational model out of an ER model.

Develop a database system using the Structured Query Language (SQL).

Formally evaluate the compliance of a database design with the main normal forms

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Databases and Database Users	1
2. Database System Concepts and Architecture	1-2
3. Data Modeling Using the Entity-Relationship (ER) Model	2-3
4. The Enhanced Entity-Relationship (EER) Model	4
5. The Relational Data Model and Relational Database Constraints	5-6
6. Relational Database Design by ER to Relational Mapping	6-7
7. The Relational Algebra and Relational Calculus	7-8-9
8. Basic SQL & More SQL: Complex Queries, Triggers, Views, and Schema Modification	9-10-11
9. Basics of Functional Dependencies and Normalization for Relational Databases	12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Quizzes	4, 8	10%
Midterm Exam	6	40%
Project	11	10%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1431	**Course Name**	Cloud Computing
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10/ Fourth Year		
3. Course Instructor	-		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Cloud Computing: Principles and Paradigms, Rajkumar Buyya, James Broberg, Andrzej M. Goscinski, ISBN: 978-0-470-88799-8, First Edition, 2011 **a. Other supplemental materials:** This is the most recent used material but it is continuously updated each time the course is taught. 1. NIST definition of cloud computing 2. Cloudonomic's The Economics of Cloud Computing 3. The Economics of Cloud Computing 4. Virtualization overview by VMWare 5. Classification of Virtualization Environment for Cloud Computing 6. AWS overview 7. White papers about IBM Bluemix and Microsoft Azure 8. MapReduce: Simplified Data Processing on Large Clusters 9. An Operating System for Multicore and Clouds: Mechanisms and Implementation 10. Variations in Performance and Scalability when Migrating n-Tier Applications to Different Clouds 11. CellCloud: A Novel Cost Effective Formation of Mobile Cloud Based on Bidding Incentives 12. MediaPaaS: a Cloud-based Media Processing Platform for Elastic Live Broadcasting		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** Cloud computing enables ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources such as networks, servers, storage, and applications that can be provisioned and managed seamlessly. In this course, students will be exposed to Cloud Computing concepts and architecture in addition to economical and technical constraints. Cloud development, management, performance, and security issues will be also addressed.		
b. Prerequisites	CS 1323		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction (course objective)

By the completion of the course the student should be able to:

This course aims to provide the fundamental concepts of Cloud Computing paradigm such as architecture, economics, technology, cloud development, management, performance, and security.

b. Course Learning Outcomes

Knowledge and Understanding:

At the end of the course, Students will be able to:

Describe Cloud Computing taxonomy, characteristics, architectures, and benefits.

List Cloud Computing building blocks and concepts (e.g. virtualizations, storage, OS, management layers).

Identify various systems constraints and performance tradeoffs when deploying and implementing Cloud solutions.

Skills:

At the end of the course, Students will be able to:

Analyze the economic feasibility of various deployments scenarios.

Develop basic distributed processing solutions for large using MapReduce technique.

Develop small applications using well-known Cloud services (e.g. AWS, Azure).

Values:

Work effectively within a team to accomplish project's objectives.

7. Brief list of topics to be covered

Chapter	**Weeks**
1. Introduction to Cloud Computing	1-2
2. Economics of Cloud Computing	3
3. Key Technologies	4
4. Public Clouds Offerings (Azure, AWS)	5-6
5. Cloud Applications	7-8
6. Performance issues in Cloud Computing	9
7. Cloud Management (Billing, SLAs)	10-11
8. Security and Privacy Issues Related to the Cloud	12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Quizzes	4, 7, 9, 11	20%
Midterm Exams	8	20%
Assignments	All semester	10%
Group Project	11	10%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1432	**Course Name**	Selected Topics in Cyber Security and Networks
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10/ Fourth Year		
3. Course Instructor	Dr. Milad Muhammad Al -Tizdante		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Selected research articles in the field of cybersecurity and computer networks Matt Bishop. Computer Security Art and Science. Addison-Wesley David F. Ferraiolo, D. Richard Kuhn, and Ramaswamy Chandramouli. Role-Based Access Control (2nd Edition), Chapters 1-2. Artech House, 2007 **a. Other supplemental materials:** None		
5. Specific course information	**b. Brief description of the content of the course (catalog description)** The course covers selected topics in the field of cybersecurity and computer networks. The course content will be continuously improved to reflect recent advances in the field of study and provide students with highly demanded relevant skills and knowledge required for industry and labor market.		
b. Prerequisites	CS 1372		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction (course objective)

By the completion of the course the student should be able to:

This course aims to provide latest trends in both industry and research relating to the areas of computer networks and security. In particular, students will have the ability to work on research related topics and understand the latest trends in industry. Also, students will learn how to work on improving current solutions to problems in both academia and industry

b. Course Learning Outcomes

Knowledge and Understanding:

Describe recent trends in research and industry in the field of cybersecurity and computer networks.

Recognize the current cybersecurity threats and potential impacts on individuals, organizations, and the society

Identify the contributions of research articles, and their academic and industrial applications

Skills:

At the end of the course, Students will be able to:

Apply state-of-the-art tools related to cybersecurity and computer networks

Use latest technologies in the field of computer networks and cybersecurity

Develop effective solutions using acquired skills related to cybersecurity networks

Values:

Work effectively in a team to solve particular problem related to the fields of cybersecurity and computer networks.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Cybersecurity threats	1-2
2. Computer Networks Technology	3-4
3. Vehicular ad-hoc networks (VANETs) technology	5-6
4. Crypto graphic Algorithms	7-8
5. Access control for industrial systems	9
6. O.S. Security	10
7. Research topics in Cyber Security	11-12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Quizzes	4, 7, 9, 11	20%
Midterm Exams	8	20%
Assignments	All semester	10%
Group Project	11	10%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1445	**Course Name**	Web Application Development
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10 / Fourth Year		
3. Course Instructor	Amal Al-Ghafis		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** Internet & World Wide Web, How To Program, Paul Deitel, Harvey Deitel, and Abbey Deitel, 5th edition, ISBN-10: 0-13-215100-6, ISBN-13: 978-0-13-215100-9, Prentice Hall, 2011. **a. Other supplemental materials:** Learning Web Design: A Beginner's Guide to XHTML, Style Sheets, and Web Graphics, Niederst, O'Reilly, 2007. Modern Full-Stack Development: Using TypeScript, React, Node.js, Webpack, and Docker 1st ed, 2020		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course explains to students Internet concepts, technologies, and tools and introduces their usage for creating web sites and developing web applications.		
b. Prerequisites	CS 1344		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

This course aims to:
Understand the typical components of the client server application paradigm.
Learn and use tools and languages such as HTML and CSS for web page design.
Learn and use tools and languages such as JavaScript for client-side scripting.
Learn and use tools and languages such as NodeJS for server-side scripting.
Get introduced to data design and exchange through internet.
Learn to work effectively in teams to develop Web based solutions.

b. Course Learning Outcomes

Knowledge and Understanding:

Understand the typical components of the client server application paradigm.
Describe usage of tools and languages for web page design, client-side scripting, server-side scripting, and data exchange.

Skills:

Design Web pages for a specific application based on standard guidelines.
Develop Web application to a specific usage based on software requirements analysis.
Clearly communicate ideas and solutions of programming problems to others
Use of word processing tools to prepare software documentation and reports.
Write software documentation effectively within a case study project.

Values:

Work effectively in teams to develop Web based solutions.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Web Basics and Concepts	1
2. HTML	2-3
3. CSS	4-5
4. JavaScript	6-7
5. NodeJS	8-9
6. MongoDB	10
	11-12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Mid Exam	8	20%
Project	10,11	20 %
Quizzes	5, 10	20 %
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1446	**Course Name**	Mobile Application Development
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10 / Fourth Year		
3. Course Instructor	Dr. Fatima Ratian Zaal Al -Anzi		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** Griffiths, D., & Griffiths, D. (2017). Headfirst Android Development: a brain-friendly guide. " O'Reilly Media, Inc." **a. Other supplemental materials:** Murphy, M. L. (2008). The busy coder's guide to Android development. United States: CommonsWare, 2008.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course introduces mobile software development for students who have Java programming skills. Students will learn to build mobile apps for phones and tablets using the Android Java SDK. Students will explore the emerging mobile ecosystem, location-aware software, and advanced programming topics including threads, sensors, APIs and databases.		
b. Prerequisites	CS 1344		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

This course aims to:

Know the evolution of modern mobile operating platforms and their operating systems

Learn the basics of creating and applying mobile applications that use a graphical user interface, event handling, persistent data storage, media processing, and communicating through the communication interfaces available on mobile devices

Develop Android apps using the right programming techniques

Be able to collaborate and work in a group on the specification and implementation of mobile applications, taking on different roles in the team on the course topics.

b. Course Learning Outcomes

Knowledge and Understanding:

Recognize the fundamentals of cross-platform mobile application development with a focus on the React Native framework (RN).

Skills:

Creating and applying mobile applications that use a graphical user interface, event handling, persistent data storage, media processing, and communicating through the communication interfaces available on mobile devices.

Develop mobile applications with useful features like accessing device hardware components and applying security measures

Values:

Collaborate and work in a group on the specification and implementation of mobile applications, taking on different roles in the team on the course topics.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Introduction to mobile platform: virtual machine, development tools, Java packages, emulators, services	1
2. Structure and lifecycle of an application for mobile system CSS.	2-3
3. Graphical User Interface: preparing containers and components, management of component layout, event handling.	4-5
4. Processing of application resources, content providers, filesystem.	6-7
5. Data persistence: backups, databases.	8-9
6. Application security and permissions: security architecture, application signing, user identification, file access, declaration and verification of permissions.	10
7. Network communication and internet applications.	11-12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Mid Exam	8	20%
Project	10,11	20 %
Quizzes	5, 10	20 %
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1447	**Course Name**	Game Application Development
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10 / Fourth Year		
3. Course Instructor	Afnan Abdul Rahman Ibrahim Al -Muammar		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** Game Design Workshop: A Playcentric Approach to Creating Innovative Games, Third Edition, 2014, Tracy Fullerton **a. Other supplemental materials:** Schell, J. (2014). The Art of Game Design: A book of lenses. AK Peters/CRC		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course teaches students effective and structured procedures used to design and develop game software. It provides students with the foundation in game design and development necessary for use in the workforce or for further study in the field of game development and creation.		
b. Prerequisites	CS 1344		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

The purpose of this course is to describe the history and evolution of video and computer games and game genres, develop Action Games, demonstrate Level Design in Game Programming, and use a programming language and game platforms to Code Game Programs.

b. Course Learning Outcomes

Knowledge and Understanding:

Describe the history and evolution of video and computer games and game genres

Skills:

Develop Action Games.

Evaluate the Level Design in Game Programming.

Use a programming language and game platforms to Code Game Programs.

Values:

Collaborate and work in a group on the specification and implementation of games applications, taking on different roles in the team on the course topics.

7. Brief list of topics to be covered

Chapter	No. Weeks
1. The Development of an Industry: How Did We Get Here?	1
2. Game Elements: Genres, Platforms & Applications. Player Elements: Market, Theory & Interactivity.	2-3
3. Scenarios – Creating Compelling Content	4-7
4. Strategy: Development & Business Cycles. Preproduction: Planning & Processing. Production: Prototyping & Play testing. Post-Production: Maintenance & Marketing.	8-11
5. Miscellaneous topics: Course Exams, Team Project, and Students Presentations	12

8. Any other information

Course Assessment

Activity	Assessment Week#	and % (Marks)
Mid Exam	8	20%
Project	10,11	20 %
Quizzes	5, 10	20 %
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1448	**Course Name**	Selected Topics in Platform based Application Development
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10 / Fourth Year		
3. Course Instructor	-		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** Selected research articles in the field of Platform based Application Development.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** The course presents essential and advanced knowledge relating to Platform based Application Development. Students will have the ability to work on research related topics and understand the latest trends in industry. They will also work on improving current solutions to problems in both academia and industry. This course is a research-based course. The topics and papers are reviewed, determined, and finalized before the beginning of each semester.		
b. Prerequisites	CS 1344		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

This course aims to: presents the students with latest trends in both industry and research relating to Platform based Application Development.

b. Course Learning Outcomes

Knowledge and Understanding:

Describe recent trends in research and industry for Platform based Application Development

Define the different approaches to develop applications

Recall application development requirements using standards and best practices

Skills:

Use recent techniques for developing a platform based application.

Values:

Work effectively in a team to solve particular problem related to the field of platform based application development

7. Brief list of topics to be covered

Chapter	No. Weeks
1. Overview of platforms (Web, Mobile, Game, Industrial etc)	1
2. Programming via platform-specific APIs	2
3. Overview of Platform Languages (Objective C, HTML5, etc)	3
4. Programming under platform constraints	4
5. Web programming languages (HTML5, Java Script, PHP, CSS, etc.)	5
6. Web platform constraints	6
7. Software as a Service (SaaS)	7
8. Mobile Programming Languages	8
9. Challenges with mobility and wireless communication	8
10. Location-aware applications	9
11. Performance / power tradeoffs	10
12. Mobile platform constraints	11
13. Emerging Technologies	12

8. Any other information

Course Assessment

Activity	Assessment Week#	and % (Marks)
Mid Exam	8	20%
Class Presentations & assignments	9	20 %
Project	11	20 %
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1462	**Course Name**	Natural Language Processing
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10 / Fourth Year		
3. Course Instructor	Dr. Amal Suleiman Muhammad Al -Saif		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Daniel Jurafsky and James H. Martin. 2009. Speech and Language Processing (2nd Edition). Prentice-Hall, Inc., USA. (The 3rd edition is coming out soon and should be adopted instead) Daniel Jurafsky and James H. Martin. 2009. Speech and Language Processing (3rd Edition). https://web.stanford.edu/~jurafsky/slp3 **a. Other supplemental materials:** Nizar Y. Habash, Introduction to Arabic Natural Language Processing, Morgan & Claypool, 2007.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course introduces the basic concepts of Natural Language Processing (NLP) including the NLP levels (i.e., morphological, lexical, syntactic, and semantic analysis levels) and the techniques and methods for addressing each one of these four levels. Examples include the methods for POS text tokenization and segmentation, POS tagging, syntactic parsing, and disambiguation. The course covers various topics of language modelling as an approach to utilize pre-trained language models for many other downstream tasks such as Information Extraction, Dialog, and text generation systems, and relating these concepts to other CS fields such as Machine & Deep Learning and AI. In addition, the course offers practical application of those concepts through assignments and projects.		
b. Prerequisites	CS 1361		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

Upon the course completion the student will be able to tackle fundamental concepts of NLP to enable the student to:
Learn about NLP classical pipeline to analyze natural language related problems.
Design and implement different NLP-based systems using machine and deep learning approaches.
Evaluate different NLP systems using appropriate evaluation measures.

b. Course Learning Outcomes

Knowledge and Understanding:

Describe fundamental concepts and tasks of natural language processing.
Understand the NLP processing pipeline at morphological, lexical, syntactic, semantic analysis level.

Skills:

Analyze natural language related problems using NLP algorithms and techniques.
Develop NLP-based systems to meet the goals of a natural language task.
Evaluate performance of different NLP-based systems.

Values:

Work effectively in teams to design and implement a program to solve an NLP task.

7. Brief list of topics to be covered

Chapter	No. Weeks
1. Introduction: NLP as an important field in AI and their current challenges and advancements (especially for Arabic Language); abstract definitions of NLP pipelines (morphological, lexical, syntactic, semantic analysis);	1
2. Morphological and Lexical Analysis Level	2-3
3. Syntactic Analysis Level	4-5
4. Semantic Analysis Level	6-7
5. Language Modelling	8-9
6. NLP Tasks and the Role of Machine and Deep Learning	10-11
7. Misc. Topics: Written Exams and Students Projects	12

8. Any other information

Course Assessment

Activity	Assessment Week#	and % (Marks)
Mid Exam	7	20%
Programming Assignments	As per preference	10 %
Short Quizzes	As per preference	10 %
Group Project	10-11	20%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1463	**Course Name**	Optimization and Metaheuristics
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10 / Fourth Year		
3. Course Instructor	Prof. Hashemi bin Ahmed bin Nasser		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** Metaheuristics from design to implementation, El-Ghazali talbi, Publisher: John Wiley & Son. 2009. Essentials of Metaheuristics A Set of Undergraduate Lecture Notes by Sean Luke, Department of Computer Science George Mason University, Second Edition Online Version 2.3, February, 2016. **a. Other supplemental materials:** Mohammad Solgi, Omid Bozorg-Haddad, Hugo Loaiciga "Meta-heuristic and Evolutionary Algorithms for Engineering Optimization", November 2017, Publisher: John Wiley & Son.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course presents an overview of the main metaheuristics used to solve hard optimization problem. Many real-life applications in several disciplines as engineering, operational research, bioinformatics, robotics, etc. involve hard optimization problems. This course aims to introduce the fundamentals of metaheuristic optimization, as well as some popular metaheuristic algorithms, such as genetic algorithms, simulated annealing, tabu search and others.		
b. Prerequisites	CS 1361		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

Upon completion of this course, students will have gained knowledge of the fundamentals of heuristic concepts and metaheuristic algorithms that enable them to: Define heuristics for specific problems, Implement the metaheuristics on various specific optimization problems. Evaluate the performance of the metaheuristics on optimization problems recognized to be difficult in the literature.

b. Course Learning Outcomes

Knowledge and Understanding:

Recognize the concepts and theory of metaheuristics.
Identify different techniques and tools of metaheuristics.

Skills:

Design metaheuristic algorithm for a specific optimization problem.
Critically analyze the performance of metaheuristics
Apply metaheuristics in solve an optimization problem.

Values:

Work effectively on teams to accomplish a common goal.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Introduction: Optimization (models, applications, and methods) and common concepts for metaheuristics.	1-2
2. Single solution-based metaheuristics: Local search, Iterative local search, Simulated Annealing, Tabu Search, Guided local search, and others	3-5
3. Population-based metaheuristics: Evolutionary Algorithms (Genetic Algorithm, Evolution Strategies, and others) and Swarm Intelligence (Ant Colony Optimization, Particle Swarm Optimization, others)	6-8
4. Advance topics: multi-objective optimization, Hybrid Metaheuristics, Parallel Metaheuristics	9-11
5. Miscellaneous topics: Course Exams, Team Project, and Students Presentations	12

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Mid Exam	8	20%
Quizzes	5,11	20 %
Group Project	As per preference	20 %
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1464	**Course Name**	Digital Image Processing
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10 / Fourth Year		
3. Course Instructor	Dr. Fahman Ali Ismail Saeed		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** R.C. Gonzales & R.E. Woods "Digital Image Processing" (2nd or 3rd edition), Prentice Hall, ISBN 0-201-18075-8. Richard Szeliski, Computer Vision: Algorithms and Applications, 2nd , 2022. **a. Other supplemental materials:** Programming Computer Vision with Python" by Jan Erik Solem. ACM transaction on Image Processing.		
8. Specific course information	**a. Brief description of the content of the course (catalog description)** This course is an introduction to image processing and Computer vision related algorithms. Image processing and computer vison is an elective course that will introduce you to an exciting topic in computer science. It should be fun and not too much of a struggle for you. Make sure that you have had the prerequisites. Depending on your level of programming experience and/or mathematics background, the course may be challenging for you. If you do not understand the material, ask for help immediately. Ask questions in class. If one student is confused about something, then maybe others are also confused and grateful that someone asked. Come and see me or the TF for help or send us email. Our task is to help you learn a very interesting topic, image analysis techniques and concepts. Areas examined include such as Imaging sensors and their principles; Image representation and storage, coding and compression techniques, lossy versus lossless; Techniques for noise reduction. Image enhancement including contrast manipulation, histogram equalization, edge highlighting; Filtering and transform techniques for image processing including two dimensional Fourier transforms, wavelets and convolution; Spatial transformations and image registration. Segmentation and thresholding techniques; Applications of morphology to image processing including erosion, dilation and hit-or-miss operations for binary and grey scale images; Image feature estimation such as edges, lines, corners, texture and simple shape measures. Object classification, template matching techniques and basic image based tracking will be also examined.		
b. Prerequisites	CS 1361		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

9. Specific goals for the course

a. Specific outcomes of instruction

In this course, you will learn the basics of Digital Image Processing. Explain the sampling, quantization, interpolation, and pixel connection concepts. Apply transformations of intensity and histogram processing. Use spatial and spectral filtering techniques. Extract useful features for different purpose.

b. Course Learning Outcomes

Knowledge and Understanding:

Show a wide knowledge of fundamental image processing and image analysis techniques and concepts (linear and non-linear filtering, morphological operators, de-noising, edge detection, compression, shape metrics and feature based recognition)

Explain the understanding and knowledge of Image Processing's core principles, and apply them to the development of algorithms.

Skills:

Analyze, demonstrate, and apply their expertise by examining image processing issues and identifying and implementing appropriate solutions.

Students will design and implement realistic solutions to a variety of typical image processing challenges and evaluate the results of their solutions, including defects, critically.

Values:

Work effectively in group project.

10. Brief list of topics to be covered

Chapter	No. Weeks
1. Digital image modeling and representation.	1
2. Image formation: scanning, digitization, sampling, quantization.	2-3
3. Morphological operators.	4-5
4. Image enhancement: intensity transformation and spatial filtering and filtering in the frequency domain.	6-7 8-9
5. Feature extraction (edge and corner detection).	10-11
6. Describing and Matching.	12
7. Miscellaneous topics: Course Exams, Team Project, and Students Presentations	

11. Any other information

Course Assessment

Activity	Assessment Week#	and % (Marks)
Mid Exam	7	20%
Quizzes	As per preference	15 %
Group Project	11	15 %
Assignments	3,10	10%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1465	**Course Name**	Neural Networks and Deep Learning
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10 / Fourth Year		
3. Course Instructor	Dr. Maha Suleiman Muhammad Al -Rabiya		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** Ian Goodfellow, Yoshua Bengio, and Aaron Courville. 2016. Deep Learning. The MIT Press. Zhang, Aston, et al. "Dive into deep learning." arXiv preprint arXiv:2106.11342 (2021). Aaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł. and Polosukhin, I., 2017. Attention is all you need. In Advances in neural information processing systems (pp. 5998-6008). **a. Other supplemental materials:** Vincent François-Lavet, Peter Henderson, Riashat Islam, Marc G. Bellemare and Joelle Pineau (2018), An Introduction to Deep Reinforcement Learning, Foundations and Trends in Machine Learning: Vol. 11, No. 3-4		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** The aim of this course is to introduce concepts of Deep Learning and its role in Artificial Intelligence. It covers the underlying concepts in machine learning. The course addresses the regularization and optimization of Deep Neural Networks (DNNs). The course also covers the anatomy of several DNN architectures and their applications, such as Convolutional Neural Networks, Sequence to Sequence Models, Attention Models, Autoencoders, Variational Autoencoders, and Deep Boltzmann Machines.		
b. Prerequisites	CS 1361		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

After completing the course, the students should be able to:
Understand ML background needed for this field.
Explain the approaches for regularization and optimization.
Know how to build several architectures of deep neural networks and identify the most appropriate ones for real world problems.

b. Course Learning Outcomes

Knowledge and Understanding:
Describe concepts of DNNs and their related background.
Identify approaches for regularization and optimization.
Explain the anatomy of different DNN architectures.
Skills:
Design appropriate DNN architectures for different tasks.
Develop, apply and test DNNs to tackle specific problems.
Values:
Work effectively in teams to design and implement a Deep Neural Network to solve a practical problem.

7. Brief list of topics to be covered

Chapter	No. Weeks
1. Introduction to ML and Neural Networks.	1
2. Deep Neural Networks	2
3. Forward and Backward passes in Networks	2
4. Convolutional Neural Networks	3
5. Deep Learning Computation (Layers and Blocks, Parameter Management,	3
6. Instantiating a Network, Custom Layers, Computing Devices)	4
7. LeNet and DenseNet Architecture	5
8. Sequence-to-Sequence Models (RNN, LSTM)	5
9. Auto encoders	6
10. Variational Auto encoders	7
11. Generative Adversarial Networks	8
12. Deep Boltzmann Machines	9
13. Regularization and Optimization	10
14. Transfer learning Architectures (AlexNet, VGG-Net, InceptionV3)	11 12
15. Project	

8. Any other information

Course Assessment

Activity	Assessment Week#	and % (Marks)
Mid Exam	8	20%
In-class application/participation	2-10	10 %
Group Project	11	30%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1466	**Course Name**	Selected Topics in Artificial Intelligence
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10 / Fourth Year		
3. Course Instructor	-		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** Russel & Norvig: Artificial Intelligence: a Modern Approach, 3rd edition, 2010. David L. Poole (Author), Alan K. Mackworth: Artificial Intelligence: Foundations of Computational Agents 1st Edition, 2010 Recent important Publications in the AI and its related applications		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This undergraduate level course introduces latest trends in industry related to the field of Artificial Intelligence (AI). The course has a mix of traditional lectures and papers/notices study about recent developments if the AI fields. The students will learn recent development of topics in a select field of AI. Professors responsible for teaching this course will continuously update its content according to recent trends in AI and the research interest of the AI research team in the department.		
b. Prerequisites	CS 1361		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

<table>
<tr><td>6. Specific goals for the course</td><td colspan="3">a. Specific outcomes of instruction
The main objectives of the course are:
Introduce students to discover recent topics in AI.
Present modern technologies and trends in the field of AI.
Assist students to develop ‘hands-on’ skills on using various tools and perform experiments and implementation in the field of Artificial Intelligence.</td></tr>
<tr><td></td><td colspan="3">b. Course Learning Outcomes
Knowledge and Understanding:
Understand a recent selected topic of Artificial Intelligence tools, concepts, algorithms, strategies, and technologies.
Recognize recent trends of industry in the field of Artificial Intelligence
Skills:
Formulate AI problems of the selected topic and design an intelligent system to solve them.
Study recent developments related to Artificial Intelligence to find out their objectives as well as academic and industrial applications.
Values:
Work effectively in a team to solve particular problem related to the fields of AI.</td></tr>
<tr><td>7. Brief list of topics to be covered</td><td colspan="2">Chapter
1. General Introduction to the selected topics in AI
2. Latest trends in industry related to the field of AI
3. Trends in Deep Learning Methodologies
4. Trends in knowledge representation and reasoning
5. Trends in AI optimization techniques
6. Trends in AI Natural Language Processing
7. Research topics in the field of AI (reading, presentation, discussions)</td><td>No. Weeks
1
2-3
4-5
6-7
8-9
10-11
12</td></tr>
<tr><td>8. Any other information</td><td colspan="3">Course Assessment</td></tr>
<tr><td></td><td>Activity
Mid Exam
Quizzes
Group Project
Final Exam
Total</td><td>Assessment Week#
7
9,4
10
Final week</td><td>and % (Marks)
20%
20 %
20%
40%
100%</td></tr>
</table>

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1471	**Course Name**	Advanced Databases
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10 / Fourth Year		
3. Course Instructor	Prof. Qaiser Abbas Muhammad Abbas		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** Ramakrishnan, R. and Gehrke, J., 2000. Database management systems. McGraw Hill Han, J., Pei, J. and Kamber, M., 2011. Data mining: concepts and techniques. Elsevier. Baeza-Yates, R. and Ribeiro, B.D.A.N., 2011. Modern information retrieval. New York: ACM Press; Harlow, England: Addison-Wesley		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** This course covers the fundamental of advanced database management systems that are essential to handle massive amount of data. The course focus on topic related to storing, accessing, and querying persistent large data. It covers in depth the use of indexes to answer efficiently queries over stored data, and data information retrieval.		
b. Prerequisites	CS 1370		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction

The main objective of this course is to provide students with the fundamentals of advanced database management system concepts and theory such as storage and file structures, classical indexes design, queries handling, and data information retrieval basics.

b. Course Learning Outcomes

Knowledge and Understanding:

Process in-depth knowledge in essential topics related to storage and files structure, indexes, and data information retrieval

Comprehend knowledge of computing and mathematics appropriately to conduct research and develop solutions in indexing and data information retrieval

Recognize leading tools & technologies, and wide range of research challenges in the field of indexing and data information retrieval

Skills:

Apply the algorithmic principles in the design and the evolution of indexes construction, and information retrieval solution.

Analyze and address research challenges in indexing, and information retrieval topics.

Communicate effectively with CS professionals in written and oral communication channels

Values:

Work effectively in a team to solve particular problem related to the fields of AI.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Introduction to Storage and Indexing	1
2. Tree based indexes & Hash based indexes	2-3
3. Other indexes (Inverted indexes, signature files, Space filling curves, Grid files), High dimensional indexing issues	4-5
4. Queries and its optimization (including range queries, k-NN queries, top N queries), Parallel and Distributed Queries	6-7
5. Introduction to Data mining (association rule mining, classification, clustering)	8-9
6. Introduction to Information retrieval (vector space model, document similarity, text search)	10-11 12
7. Emerging Trends in Databases	

8. Any other information

Course Assessment

Activity	**Assessment Week#**	**and % (Marks)**
Mid Exam	8	20%
Assignments	3	20 %
Project	6	20%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1474	**Course Name**	Networks Security
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10/ Fourth Year		
3. Course Instructor	Dr. Firas Abdulaziz AlJumaa		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Network Security: Private Communication in a Public World, Charlie Kaufman, Radia Perlman, Mike Speciner, Second Edition, Prentice Hall, 2002. Network Security Essentials: Applications and Standards, William Stallings, 6th edition, Pearson, 2017 **a. Other supplemental materials:** None		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** Network security is an important course to secure network protocols and its application. With the growth of network cyber-attacks students has to grasp concepts and sharpen their skills to apply practices of securing network applications.		
b. Prerequisites	CS 1372		
Co-requisites	No Co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction (course objective)

By the completion of the course the student should be able to:

This course aims to provide an academic overview of network security covering its main domains with more focus on formal modelling, analyzing, detecting, and mitigating network attacks. By completion of this course, students should appreciate the significance of network security in the IT realm in general and be able to demonstrate in depth knowledge in network attacks and how defend IT systems in specific.

b. Course Learning Outcomes

Knowledge and Understanding:

At the end of the course, Students will be able to:

Describe virtues and limitations of security technologies at each layer of the network stack.

Discuss how cyberattacks can be identified and mitigated in different medium such as Wireless, and web attacks

Explain how public key infrastructure supports digital signing and encryption and discuss the imitations/vulnerabilities

Skills:

At the end of the course, Students will be able to:

Evaluate the cost and tradeoffs associated with designing security protocols

Analyze and detect system and network vulnerabilities

Develop effective solutions using acquired skills related to network security

Values: Work effectively in a team to solve particular problem related to the field of network security.

7. Brief list of topics to be covered

Chapter	**No. Weeks**
1. Introduction to Network Security and Design and analysis of security protocols	1
2. Cryptographic Protocols and Network security policies.	2
3. IPSEC	3
4. PGP – S/MIME	4
5. SSH	5
6. DNSSEC	6
7. Public key infrastructure	7
8. Wireless Network Security and Mobile Cellular Networks Security	8
9. Network Infrastructure Attacks	9
10. Wireless Network Attacks and Web Application Attacks	10
11. Spam and Phishing Attacks and Vulnerability Scanning	11
12. Network security assessment	12

8. Any other information

Course Assessment

Activity	**Assessment Week# and % (Marks)**
Quizzes	As per preference 20%
Midterm Exams	As per preference 20%
Group Project	11th 20%
Final Exam	Final week 40%
Total	100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1475	**Course Name**	Software Security
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10/ Fourth Year		
3. Course Instructor	Dr. Mohammed Abdullah Al -Jamaz		
Tutorial/Practical Instructor	L; T; P		
4. Textbook, title, author, and year	**Textbook:** Du, Wenliang (2019). Computer & Internet Security: A hands-on approach. McGraw, Gary. (2013). Software security: Building security in. Addison-Wesley. **a. Other supplemental materials:** None		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** The course details secure programming techniques to defend against software vulnerabilities such as buffer overflow and code injection. It also covers how to avoid mistakes and pitfalls when using cryptographic libraries. The course also explores secure software development through the use of secure design, secure coding, program analysis, and advanced testing.		
b. Prerequisites	CS 1372 CS 1251		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction (course objective)

By the completion of the course the student should be able to:

The course details software security vulnerabilities and techniques to avoid them.

b. Course Learning Outcomes

Knowledge and Understanding

Describe an advanced secure software development framework and its applications for real world software design.

Comprehend advanced software security vulnerabilities and how to defend against them.

Skills

Apply secure software development processes to real world software development.

Implement security vulnerabilities, and their associated countermeasures.

Utilize various tools to help secure code.

7. Brief list of topics to be covered

Chapter	No. Weeks
1. Introduction to Software Security	1
2. Defining Software Security fundamentals	1-2
3. Risk Management Framework	2-3
4. Introduction to Software Security Touchpoints	4
5. Buffer Overflow Attack	5-6
6. Code Review with a Tool	6-7
7. Architectural Risk Analysis	7-8
8. Format String Vulnerability	8-9
9. Software Penetration Testing	10
10. Risk-Based Security Testing	11
11. Libc Attack	12

8. Any other information

Course Assessment

Activity	Assessment Week#	and % (Marks)
Midterm	7	30%
Assignment1	6	10%
Assignment2	9	10%
Assignment3	11	10%
Final Exam	12	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1476	**Course Name**	Digital Forensics
2. Credit hours	3	**Contact hours**	45
Level / Year	Level 10/ Fourth Year		
3. Course Instructor	-		
Tutorial/Practical Instructor	L; T		
4. Textbook, title, author, and year	**Textbook:** Guide to Computer Forensics and Investigations: Processing Digital Evidence, by Bill Nelson, Amelia Phillips, Christopher Steuart, Cengage Learning, Inc, 2019 Practical Guide to Digital Forensics Investigations, A, by Darren R. Hayes, Pearson IT Certification, 2021 **a. Other supplemental materials:** Technical Reports.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** Digital forensics is the process of collecting, analyzing, and preserving electronic data to investigate and solve crimes involving digital devices or networks. This course provides a broad introduction to the digital forensics profession, the investigator's office and laboratory, data acquisition, Windows forensics, Linux forensics, processing crime and incident scenes, recovering graphics files, cloud forensics, digital forensics analysis and validation, E-mail and social media investigations, mobile device forensics and the internet of anything, virtual machine forensics, network forensics, current digital forensics tools, report writing for high-tech investigations, expert testimony in digital investigations and ethics for the expert witness.		
b. Prerequisites	CS 1372		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Elective		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction (course objective)

By the completion of the course the student should be able to:

Upon completion of this course, students will have gained knowledge of the fundamentals of digital forensics concepts and theory that enable them to apply them on real use cases.

b. Course Learning Outcomes

Knowledge and Understanding

Recognize the concepts and theories of digital forensics.

Identify different techniques and tools of digital forensics.

Skills

Apply digital forensics methods

Utilize various digital forensics tools

Values

Work effectively on teams to accomplish a common goal.

Realize professional, ethical, legal, security and social issues and responsibilities

7. Brief list of topics to be covered

Chapter	No. Weeks
1. Understanding the Digital Forensics Profession and Investigations and the Investigator's Office and Laboratory	1
2. Data Acquisition	2
3. Processing Crime and Incident Scenes	2-3
4. Working with Windows and CLI Systems	3-4
5. Linux and Macintosh File Systems	4
6. Recovering Graphics Files	5
7. Mobile Device Forensics and The Internet of Anything	5-6
8. Cloud Forensics, E-mail and Social Media Investigations	6-7
9. Virtual Machine Forensics, Live Acquisitions, and Network Forensics	7-8
10. Digital Forensics Analysis and Validation	8
11. Current Digital Forensics Tools	9
12. Report Writing for High-Tech Investigations	10
13. Expert Testimony in Digital Investigations and Ethics for the Expert Witness	10-11
14. Miscellaneous topics: Course Exams, Team Project, and Students Presentations	12

8. Any other information

Course Assessment

Activity	Assessment Week#	and % (Marks)
Midterm	8	20%
Assignments	As per preference	25%
Quizzes	As per preference	15%
Final Exam	Final week	40%
Total		100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1481	**Course Name**	Professional Development Seminar
2. Credit hours	2	**Contact hours**	30
Level / Year	Level 10/ Fourth Year		
3. Course Instructor	Dr. Lamis Abdullah Suleiman Al -Hazaa		
Tutorial/Practical Instructor	L		
4. Textbook, title, author, and year	**Textbook:** F Day, Robert A., and Gastel, B., How to Write and Publish a Scientific Paper., 7th edition, Greenwood Publishing Group. 2012 **a. Other supplemental materials:** A Manual for Writers of Research Papers, Theses, and Dissertations, Kate L. Turabian, 8th edition, 2013.		
5. Specific course information	**a. Brief description of the content of the course (catalog description)** The course format is face-to-face weekly class meetings with face-to-face class discussions. Most class meetings include a short instructor lecture of no more than 10-15 minutes followed by small group topic discussions, consisting of groups of no more than 4-5 students. Additionally, the course utilizes group discussions, a group research project, a group research presentation and a unit on preparing for professional interviews which includes a unit on technical resume preparation and technical and situational interview preparation. Group Discussions Group Research Project and Presentation Professional Interviews Preparation		
b. Prerequisites	CS 1352 CS 1361 CS 1344		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

6. Specific goals for the course

a. Specific outcomes of instruction (course objective)

By the completion of the course the student should be able to:

Move the content of CS492 to this course. Emphasize group work and group discussion during lecture. Enhance students written and oral communication skills through practicing face to face professional interviews, writing their professional resume and applying for their summer internships.

b. Course Learning Outcomes

Knowledge and Understanding:

Describe how to avoid plagiarism

Skills:

Analyze scientific papers.

Write a technical resume in a professional way.

Write a scientific proposal for a CS graduation project in a professional way.

Write the problem statement, the aim and objectives, the background and the literature review of the proposed project in a professional way.

Values

Operate effectively among a team during the process of choosing the topic and writing the proposal and the literature review.

Perform a successful professional technical interview related to computer science domain.

Demonstrate project problem definition, aim and objectives, background and literature review in a professional way.

7. Brief list of topics to be covered

Chapter	No. Weeks
1. Dynamics of oral, written, and electronic team and group communication	1
2. Writing professional resume	2
3. Reading, understanding and summarizing research papers	3-4
4. Plagiarism & Intellectual Property	4-5
5. Citing and referencing academic sources	5-6
6. How to write project proposal	6-7
7. How to conduct a research (formulating the problem statement and performing a literature review)	7-8
8. How to write the Introduction chapter	8-9
9. How to write Literature Review chapter	9-10
10. Designing and presenting a project (format and skills)	10
11. Designing and presenting a project poster (format and skills)	11
12. Public presentations of group projects	12

8. Any other information

Course Assessment

Activity	Assessment Week# and % (Marks)	
Group Research Activity Summarize and present review paper	3	10%
Group Research Activity Summarize and present review paper	4	10%
Group Research Activity Summarize and present review paper	6	15%
Group Research Activity Summarize and present review paper	8	15%

	Technical Resume Development (Single) Project -Review Paper Development and presentation (group) Total	9 10-11	10% 40% 100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1490	**Course Name**	Practical Training 1
2. Credit hours	8	**Contact hours**	120
Level / Year	Level 11/ Fourth Year		
Tutorial/Practical Instructor	P		
3. Textbook, title, author, and year	**Textbook:** None **a. Other supplemental materials:**		
4. Specific course information	**a. Brief description of the content of the course (catalog description)**		
b. Prerequisites	CS 1481 CS 1343 CS 1323		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

<table>
<tr><td rowspan="2">5. Specific goals for the course</td><td colspan="2">a. Specific outcomes of instruction (course objective)</td></tr>
<tr><td colspan="2">b. Course Learning Outcomes
Knowledge and Understanding
Recognize the computing-related techniques, skills, and tools necessary to find solutions in the workplace
Skills:
Develop computing solutions using appropriate skills and techniques
Analyze problems and identify efficient solutions in the field experience environment

Communicate effectively with a range of stakeholders in the work environment
Values:
Work effectively on teams or lead teams to accomplish tasks in the field of computing

Recognize the importance of continuous professional development
Demonstrate the ability to act ethically, responsibly, and professionally in the work environment</td></tr>
<tr><td>6. Brief list of topics to be covered</td><td colspan="2"></td></tr>
<tr><td>7. Any other information</td><td colspan="2">Course Assessment</td></tr>
<tr><td></td><td>Activity
Weekly Tasks/Activities Report
Final Trainee Evaluation
Attendance Report
Final Training Report
Total</td><td>Assessment Week# and % (Marks)
Every week 25%
At the end of the training period 25%
At the end of the training period 10%
At the end of the training period 40%
100%</td></tr>
</table>

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS1491	**Course Name**	Practical Training 2
2. Credit hours	8	**Contact hours**	120
Level / Year	Level 12/ Fourth Year		
Tutorial/Practical Instructor	P		
3. Textbook, title, author, and year	**Textbook:** None **a. Other supplemental materials:**		
4. Specific course information	**c. Brief description of the content of the course (catalog description)**		
b. Prerequisites	CS 1490		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

<table>
<tr><td rowspan="2">5. Specific goals for the course</td><td colspan="2">a. Specific outcomes of instruction (course objective)</td></tr>
<tr><td colspan="2">b. Course Learning Outcomes
Knowledge and Understanding
Recognize the impact of computing on organizations, and the society in the context of field experience environment
Skills:
Develop efficient computing solutions using innovative methods and advanced skills
Analyze problems using logical thinking techniques and identify efficient solutions in the work environment
Implement computer based system or components in an effective way to meet desired needs
Communicate effectively with a range of stakeholders in the work environment
Values:
Work effectively on teams or lead teams to accomplish a common goal
Recognize the importance of continuous professional development
Demonstrate the ability to act ethically, responsibly, and professionally in the work environment</td></tr>
<tr><td>6. Brief list of topics to be covered</td><td colspan="2"></td></tr>
<tr><td>7. Any other information</td><td colspan="2">Course Assessment</td></tr>
<tr><td></td><td>Activity
Weekly Tasks/Activities Report
Final Trainee Evaluation
Attendance Report
Final Training Report
Total</td><td>Assessment Week# and % (Marks)
Every week 25%
At the end of the training period 25%
At the end of the training period 10%
At the end of the training period 40%
100%</td></tr>
</table>

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1492	**Course Name**	Graduation Project 1
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 11/ Fourth Year		
Tutorial/Practical Instructor	P		
3. Textbook, title, author, and year	**Textbook:** None **a. Other supplemental materials:**		
4. Specific course information	**a. Brief description of the content of the course (catalog description)** The project is a significant computer science project requiring extensive research and development, conducted under the general guidance of an approved faculty member, and conforming to departmental project guidelines. The course will allow the student to choose a topic from many computer science areas and an advisor so that he can continue his project in the next project course.		
b. Prerequisites	CS 1481 CS 1343 CS 1323		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

5. Specific goals for the course	**a. Specific outcomes of instruction (course objective)** By the completion of the course the student should be able to: The main purpose of the course is to guide students through their graduation projects by dividing students into groups, help them to choose a project topic and then assigning supervisors to them. Supervisors continue to guide student through a weekly two hours meeting augmented with emails and other types of communications following the department graduation project guidelines.	
	b. Course Learning Outcomes **Knowledge and Understanding** Recognize the concepts of computing and mathematics appropriate to the discipline Identify the current techniques, skills, and tools necessary for the computing practice Recognize local and global impact of computing on individuals, organizations, and the society **Skills:** Analyze scientific papers and write a literature review Analyze the software system requirements. Communicate effectively with CS professional using written methods **Values:** Operate effectively among a team and take responsibility Recognize the importance of pursuing higher degrees and participating in professional development activities. Identify professional, ethical, and legal, security and social issues and responsibilities related to the development of computing solutions.	
6. Brief list of topics to be covered	The course does not offer any topics, only important guidelines, materials and milestones are given to the students and supervisors by the course coordinator. Meetings and other forms of communication with supervisors. (all weeks)	
7. Any other information	**Course Assessment**	
	Activity	**Assessment Week# and % (Marks)**
	Project proposal (Course Coordinator)	2 5%
	Originality reports (Course Coordinator)	12 3%
	Following Directions (Course Coordinator)	All 2%
	Project report (Supervisor)	12 38%
	Interpersonal skills & responsibilities (Supervisor)	All 12%
	Project report (Examiners)	13 30%
	Presentation (Examiners)	13 10%
	Total	100%

Computer Science Department

COURSE DESCRIPTION
Third Semester, 2023/2024

1. Course Number	CS 1493	**Course Name**	Graduation Project 2
2. Credit hours	4	**Contact hours**	60
Level / Year	Level 12/ Fourth Year		
Tutorial/Practical Instructor	P		
3. Textbook, title, author, and year	**Textbook:** None **a. Other supplemental materials:**		
4. Specific course information	**d. Brief description of the content of the course (catalog description)** The project is a significant computer significant computer science project requiring extensive research and development, conducted under the general guidance of an approved faculty member, and conforming to departmental project guidelines. The course will allow the student to complete "CS 1492: Graduate Project 1" course from previous semester. As its name suggests "Graduate Project 2" is a semester long assignment chosen by a student or a group of students on a topic related to the CS curriculum and involves out-of-class research and a development of a computer system. The course is designed to make a comprehensive assessment of the skills the students is expected to have once graduated and how well he is equipped with the know-how, self-discipline, and attitudes required for a typical professional environment. The final report should objectively and clearly discuss the problem statement, the system requirements, the potential solutions and/or approaches, arguments about the choice of the solution.		
b. Prerequisites	CS 1492		
Co-requisites	No co-requisites		
c. Required, Elective, or Selected elective	Required		

L Lecture; T Tutorial; P Practical

5. Specific goals for the course	**a. Specific outcomes of instruction (course objective)** By the completion of the course the student should be able to: The main purpose of CS 1493 is to put in practice and assess the ability of the student to use the skills and to apply knowledge he\she acquired throughout the curriculum in a real-world challenge. In addition, the students can learn and use some of the new technologies recently available in their projects. In sum, the students can use what they learned during their studies along with some new technologies in market to let them ready for the employment.	
	b. Course Learning Outcomes **Knowledge and Understanding** Recognize the concepts of software systems design. Identify techniques, and tools required to develop software systems. Explain positive and negative impacts of a computing solution on individuals, organizations, and or the society. **Skills:** Analyze a problem and define the computing requirements appropriate to its solution. Apply software development principles to define the computing requirements appropriate to solve a computing problem. Design a computer-based system, component, or program to meet desired needs Communicate the project progress and findings effectively with CS professionals using oral presentation. Develop a computer based system, component, or program to meet desired needs using design, implementation and evaluation principles. Apply computer science theory in the modeling and design of computer-based systems **Values:** Work effectively on teams to accomplish a common goal Recognize the need for and an ability to engage in continuing professional development. Realize professional, ethical, legal, security and social issues and responsibilities	
6. Brief list of topics to be covered	The course does not offer any topics, only important guidelines, materials and milestones are given to the students and supervisors by the course coordinator. Meetings and other forms of communication with supervisors. (all weeks)	
7. Any other information	**Course Assessment**	
	Activity	**Assessment Week# and % (Marks)**
	Originality reports (Coordinator)	12 6%
	Following Directions (Course Coordinator)	All 4%
	Project report (Supervisor)	
	Programming and Research	12 20%
	Skills (Supervisor)	All 20%
	Interpersonal skills & responsibilities (Supervisor)	All 10%
	Project report (Examiners)	13 30%
	Presentation (Examiners)	13 10%
	Total	100%
